



Interactive Animations and Games

Lesson 7

Mini-Project: Robot Face

Warm Up



Journal Prompt:

What are some ways we can use variables with random numbers to make our programs different each time we run them?

Question of the Day

How can we use shapes, variables, and randomness to express our creativity?

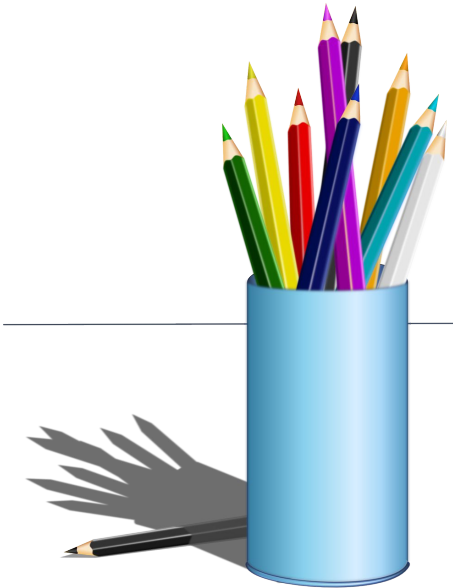
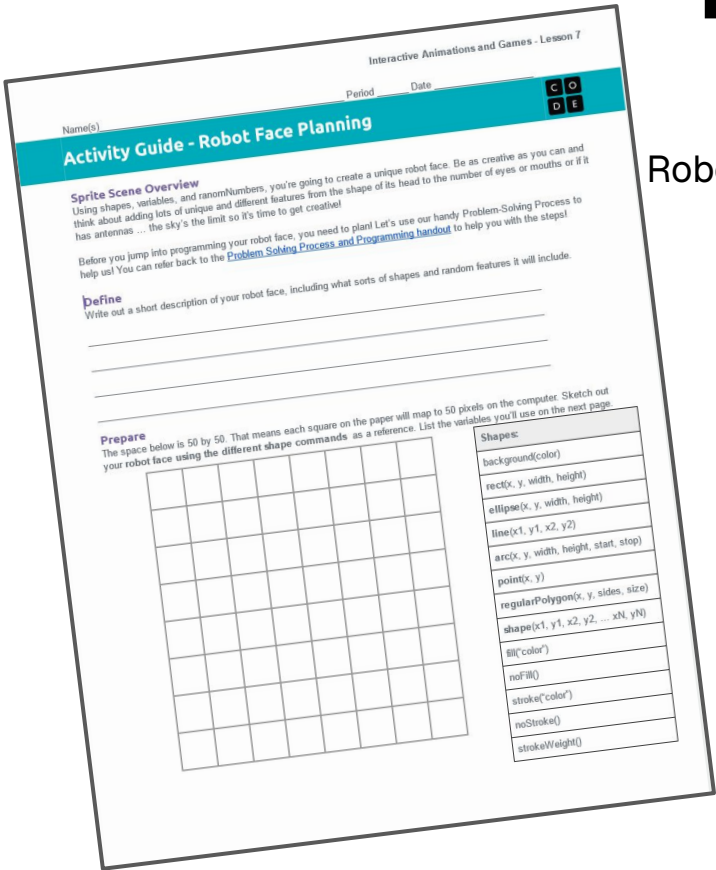
Activity





Distribute

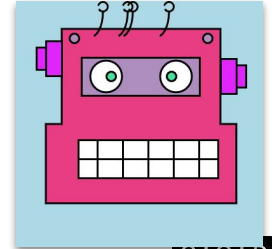
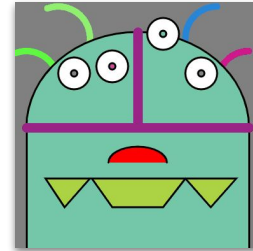
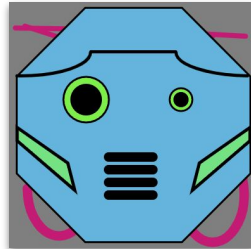
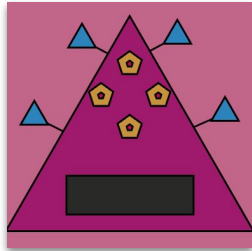
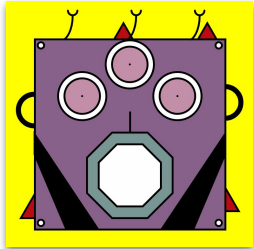
You should have:
Robot Face Planning Activity Guide
Pen/Pencil



Robot Face Planning

Your robot face can be as creative as you want!

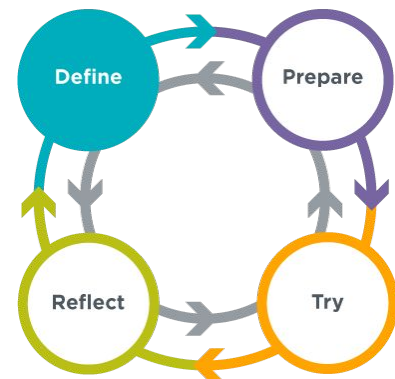
You can add lots of unique and different features from the shape of its head to the number of eyes or mouths or if it has antennas ... the sky's the limit so it's time to get creative!



Robot Faces

Your robot face will show that you can...

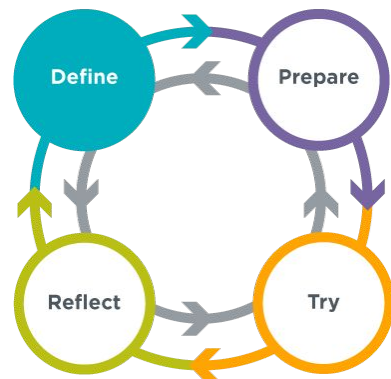
- Apply shapes, variables, and randomNumber concepts to create a program
- Use a structured process to plan and develop a program



Define

- Describe the robot face you want to create.
 - Make sure you think about features that could be random

Robot Faces

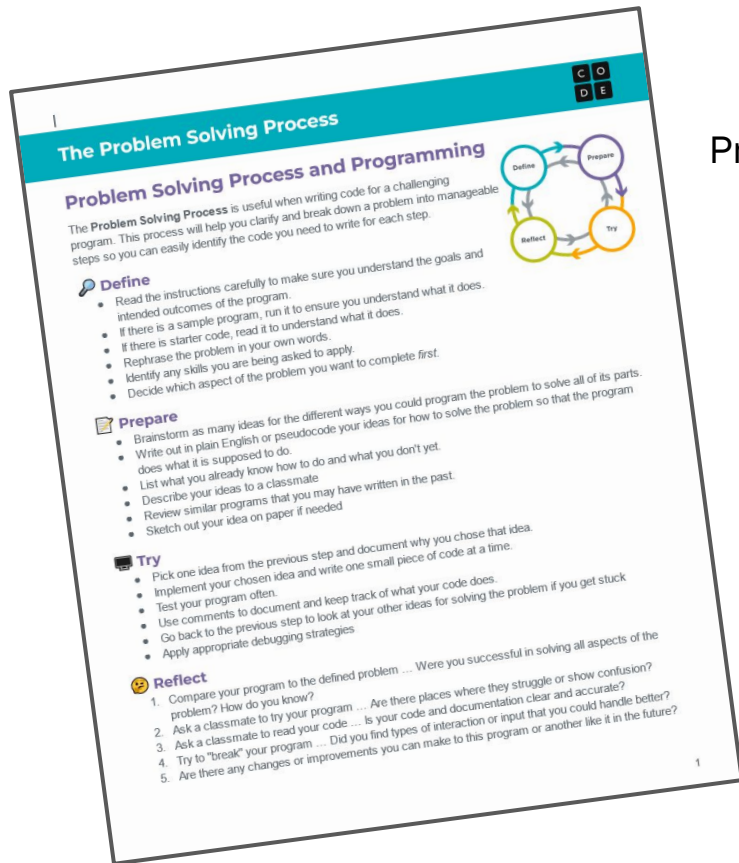
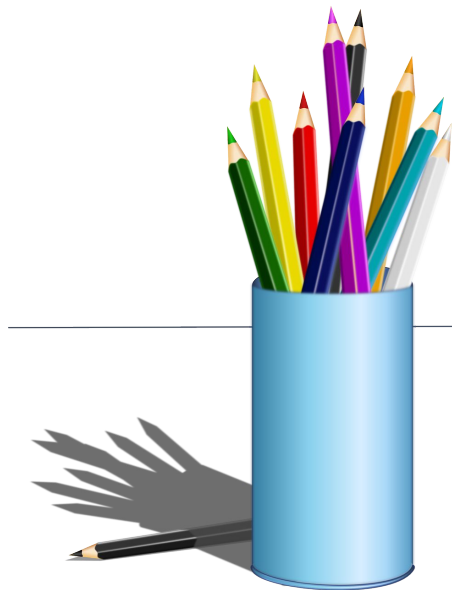


Prepare

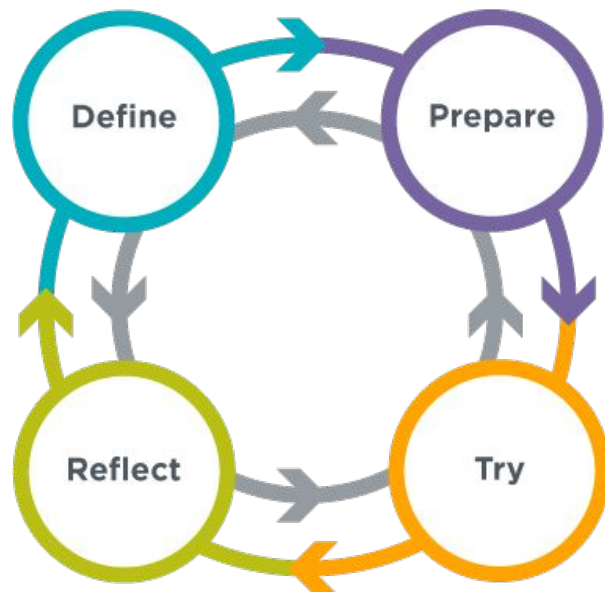
- Make a sketch of your robot face
 - Don't forget about the additional shapes you can use
 - `arc`, `line`, `shape`, `regularPolygon`, `point`
- List the variables you'll use to give your robot face random features

Distribute

You should have:
Problem Solving Process and
Programming Resource
Pen/Pencil



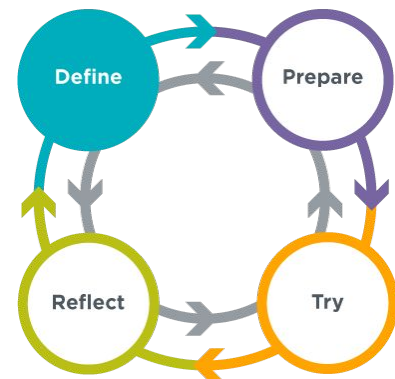
The Problem Solving Process



Robot Faces

Try

- Log into Code Studio
- If you want a review of the shape blocks from previous challenge levels, go to Lesson 7 - Level 1 and review any content you would like
- Once you feel ready, start coding your robot face in Lesson 7 - Level 2

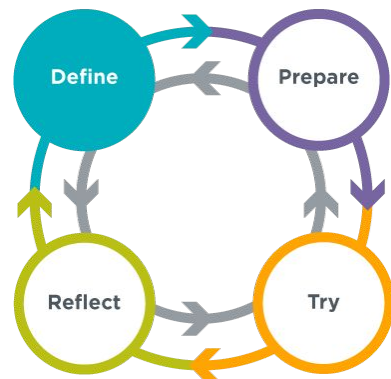


Robot Faces

Reflect

Check over your program one last time to make sure you have included everything that you want.

- Were you successful in solving all aspects of the problem?
- What part of your animation are you most proud of?
- What was the most challenging?
- Are there any changes or improvements you can make to this program or another like it in the future?





Robot Faces

Gallery Walk

Look at the robot faces your classmates have created.

- What is interesting about each creation?
- How did they use their coding skills creatively?
- Is there anything you see that you'd like to try?

Wrap Up





Question of the Day

How can we use shapes, variables, and randomness to express our creativity?



Journal Prompt:

What was one especially creative way you saw someone else use the blocks today?